

EBA0201 – PRECALCULUS

Continuation of Lecture 1: LIMITS

Part 1: Direct Substitution

Key Idea

For many functions, a limit can be found by direct substitution if the function is continuous at the given point.

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Example 1

Evaluate:

$$\lim_{x \rightarrow 3} (2x^2 - 5x + 4)$$

Solution:

$$\begin{aligned}\lim_{x \rightarrow 3} (2x^2 - 5x + 4) &= 2(3)^2 - 5(3) + 4 \\ &= 18 - 15 + 4\end{aligned}$$

Therefore:

$$\lim_{x \rightarrow 3} (2x^2 - 5x + 4) = \boxed{7}$$

Example 2

Evaluate:

$$\lim_{x \rightarrow -2} (x^3 + 4x^2 - 3)$$

Solution:

$$\begin{aligned}\lim_{x \rightarrow -2} (x^3 + 4x^2 - 3) &= (-2)^3 + 4(-2)^2 - 3 \\ &= -8 + 16 - 3\end{aligned}$$

Therefore:

$$\lim_{x \rightarrow -2} (x^3 + 4x^2 - 3) = \boxed{5}$$

Example 3

Evaluate:

$$\lim_{x \rightarrow 4} \sqrt{x + 5}$$

Solution:

$$\begin{aligned}\lim_{x \rightarrow 4} \sqrt{x + 5} &= \sqrt{4 + 5} \\ &= \sqrt{9}\end{aligned}$$

Therefore:

$$\lim_{x \rightarrow 4} \sqrt{x + 5} = \boxed{3}$$

Example 4

Evaluate:

$$\lim_{x \rightarrow 2} (3x^3 - x^2 + 6)$$

Solution:

$$\begin{aligned}\lim_{x \rightarrow 2} (3x^3 - x^2 + 6) &= 3(2)^3 - (2)^2 + 6 \\ &= 24 - 4 + 6\end{aligned}$$

Therefore:

$$\lim_{x \rightarrow 2} (3x^3 - x^2 + 6) = \boxed{26}$$

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Continuation of Lecture 1: LIMITS

Part 2: More Direct Substitution and Quadratic Factorization

Key Idea: If direct substitution gives a finite value, stop. If it gives $0/0$ and the expression is quadratic, factor and cancel the common factor.

Example 5

Evaluate:

$$\lim_{x \rightarrow 1} \frac{x^2 + 4x + 3}{x + 2}$$

Solution:

Since direct substitution gives a finite value, evaluate the expression at $x = 1$:

$$\lim_{x \rightarrow 1} \frac{x^2 + 4x + 3}{x + 2} = \frac{1^2 + 4(1) + 3}{1 + 2} = \frac{8}{3}.$$

Therefore:

$$\lim_{x \rightarrow 1} \frac{x^2 + 4x + 3}{x + 2} = \frac{8}{3}.$$

Example 6

Evaluate:

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}.$$

Solution:

Factor the numerator and cancel the common factor:

$$\begin{aligned} x^2 - 9 &= (x - 3)(x + 3), \\ \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} &= \lim_{x \rightarrow 3} \frac{(x - 3)(x + 3)}{x - 3} \\ &= \lim_{x \rightarrow 3} (x + 3) = 3 + 3 = 6. \end{aligned}$$

Therefore:

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = 6.$$

Example 7

Evaluate:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - x - 2}.$$

Solution:

Factor the numerator and denominator:

$$\begin{aligned} x^2 - 4 &= (x - 2)(x + 2), \\ x^2 - x - 2 &= (x - 2)(x + 1), \\ \lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - x - 2} &= \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{(x - 2)(x + 1)} \\ &= \lim_{x \rightarrow 2} \frac{x + 2}{x + 1} = \frac{2 + 2}{2 + 1} = \frac{4}{3}. \end{aligned}$$

Therefore:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - x - 2} = \frac{4}{3}.$$

Example 8

Evaluate:

$$\lim_{x \rightarrow 5} \frac{x^2 - 25}{x^2 - 3x - 10}.$$

Solution:

Factor the numerator and denominator:

$$\begin{aligned} x^2 - 25 &= (x - 5)(x + 5), \\ x^2 - 3x - 10 &= (x - 5)(x + 2), \\ \lim_{x \rightarrow 5} \frac{x^2 - 25}{x^2 - 3x - 10} &= \lim_{x \rightarrow 5} \frac{(x - 5)(x + 5)}{(x - 5)(x + 2)} \\ &= \lim_{x \rightarrow 5} \frac{x + 5}{x + 2} = \frac{5 + 5}{5 + 2} = \frac{10}{7}. \end{aligned}$$

Therefore:

$$\lim_{x \rightarrow 5} \frac{x^2 - 25}{x^2 - 3x - 10} = \frac{10}{7}.$$

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Part 3: Quadratic Factorization and Conjugates

Use only factorization of degree 2 expressions. Do not use cubic factorization.

Example 9

Evaluate: $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x^2 - 5x + 4}.$

Solution:

$$\begin{aligned}x^2 - 16 &= (x - 4)(x + 4) \\x^2 - 5x + 4 &= (x - 4)(x - 1)\end{aligned}$$

so

$$\begin{aligned}\lim_{x \rightarrow 4} \frac{x^2 - 16}{x^2 - 5x + 4} &= \lim_{x \rightarrow 4} \frac{(x - 4)(x + 4)}{(x - 4)(x - 1)} \\&= \lim_{x \rightarrow 4} \frac{x + 4}{x - 1} \\&= \frac{4 + 4}{4 - 1} \\&= \frac{8}{3}\end{aligned}$$

Therefore:

$$\frac{8}{3}$$

Example 10

Evaluate: $\lim_{x \rightarrow -3} \frac{x^2 - 9}{x^2 + 2x - 3}.$

Solution:

$$\begin{aligned}x^2 - 9 &= (x - 3)(x + 3) \\x^2 + 2x - 3 &= (x + 3)(x - 1)\end{aligned}$$

so

$$\begin{aligned}\lim_{x \rightarrow -3} \frac{x^2 - 9}{x^2 + 2x - 3} &= \lim_{x \rightarrow -3} \frac{(x - 3)(x + 3)}{(x + 3)(x - 1)} \\&= \lim_{x \rightarrow -3} \frac{x - 3}{x - 1} \\&= \frac{-3 - 3}{-3 - 1} \\&= \frac{3}{2}\end{aligned}$$

Therefore:

$$\frac{3}{2}$$

Example 11

Evaluate: $\lim_{x \rightarrow 5} \frac{\sqrt{x + 4} - 3}{x - 5}.$

Solution: Multiply by the conjugate:

$$\begin{aligned}\lim_{x \rightarrow 5} \frac{\sqrt{x + 4} - 3}{x - 5} &\cdot \frac{\sqrt{x + 4} + 3}{\sqrt{x + 4} + 3} \\&= \lim_{x \rightarrow 5} \frac{(x + 4) - 9}{(x - 5)(\sqrt{x + 4} + 3)} \\&= \lim_{x \rightarrow 5} \frac{x - 5}{(x - 5)(\sqrt{x + 4} + 3)} \\&= \lim_{x \rightarrow 5} \frac{1}{\sqrt{x + 4} + 3} \\&= \frac{1}{\sqrt{5 + 4} + 3} \\&= \frac{1}{3 + 3} \\&= \frac{1}{6}\end{aligned}$$

Therefore:

$$\frac{1}{6}$$

Example 12

Evaluate: $\lim_{x \rightarrow 0} \frac{\sqrt{x + 9} - 3}{x}.$

Solution: Multiply by the conjugate:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sqrt{x + 9} - 3}{x} &\cdot \frac{\sqrt{x + 9} + 3}{\sqrt{x + 9} + 3} \\&= \lim_{x \rightarrow 0} \frac{(x + 9) - 9}{x(\sqrt{x + 9} + 3)} \\&= \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{x + 9} + 3)} \\&= \lim_{x \rightarrow 0} \frac{1}{\sqrt{x + 9} + 3} \\&= \frac{1}{\sqrt{0 + 9} + 3} \\&= \frac{1}{3 + 3} \\&= \frac{1}{6}\end{aligned}$$

Therefore:

$$\frac{1}{6}$$

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Continuation of Lecture 1: LIMITS

Part 4: Limits Using the Conjugate

Key Idea: When radicals are present and direct substitution gives $0/0$, multiply by the conjugate.

$$(\sqrt{A} - B)(\sqrt{A} + B) = A - B^2$$

Example 13

Evaluate:

$$\lim_{x \rightarrow 7} \frac{\sqrt{x+2} - 3}{x - 7}$$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 7} \frac{\sqrt{x+2} - 3}{x - 7} &= \lim_{x \rightarrow 7} \frac{(\sqrt{x+2} - 3)(\sqrt{x+2} + 3)}{(x - 7)(\sqrt{x+2} + 3)} \\ &= \lim_{x \rightarrow 7} \frac{x + 2 - 9}{(x - 7)(\sqrt{x+2} + 3)} \\ &= \lim_{x \rightarrow 7} \frac{x - 7}{(x - 7)(\sqrt{x+2} + 3)} \\ &= \lim_{x \rightarrow 7} \frac{1}{\sqrt{x+2} + 3} \end{aligned}$$

Therefore:

$$\lim_{x \rightarrow 7} \frac{\sqrt{x+2} - 3}{x - 7} = \boxed{\frac{1}{6}}$$

Example 14

Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x} &= \lim_{x \rightarrow 0} \frac{(\sqrt{x+4} - 2)(\sqrt{x+4} + 2)}{x(\sqrt{x+4} + 2)} \\ &= \lim_{x \rightarrow 0} \frac{x + 4 - 4}{x(\sqrt{x+4} + 2)} \\ &= \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{x+4} + 2)} \\ &= \lim_{x \rightarrow 0} \frac{1}{\sqrt{x+4} + 2} \end{aligned}$$

Therefore:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x} = \boxed{\frac{1}{4}}$$

Example 15

Evaluate:

$$\lim_{x \rightarrow 1} \frac{\sqrt{x+8} - 3}{x - 1}$$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{\sqrt{x+8} - 3}{x - 1} &= \lim_{x \rightarrow 1} \frac{(\sqrt{x+8} - 3)(\sqrt{x+8} + 3)}{(x - 1)(\sqrt{x+8} + 3)} \\ &= \lim_{x \rightarrow 1} \frac{x + 8 - 9}{(x - 1)(\sqrt{x+8} + 3)} \\ &= \lim_{x \rightarrow 1} \frac{x - 1}{(x - 1)(\sqrt{x+8} + 3)} \\ &= \lim_{x \rightarrow 1} \frac{1}{\sqrt{x+8} + 3} \end{aligned}$$

Therefore:

$$\lim_{x \rightarrow 1} \frac{\sqrt{x+8} - 3}{x - 1} = \boxed{\frac{1}{6}}$$

Example 16

Evaluate:

$$\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9}$$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9} &= \lim_{x \rightarrow 9} \frac{(\sqrt{x} - 3)(\sqrt{x} + 3)}{(x - 9)(\sqrt{x} + 3)} \\ &= \lim_{x \rightarrow 9} \frac{x - 9}{(x - 9)(\sqrt{x} + 3)} \\ &= \lim_{x \rightarrow 9} \frac{1}{\sqrt{x} + 3} \end{aligned}$$

Therefore:

$$\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9} = \boxed{\frac{1}{6}}$$

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Continuation of Lecture 1: LIMITS

Part 5: A Useful Power Formula

For limits of the form $\lim_{x \rightarrow a} \frac{x^n - a^n}{x^m - a^m}$, the result is $\frac{n}{m} a^{n-m}$.

Example 17

Evaluate:

$$\lim_{x \rightarrow 2} \frac{x^5 - 32}{x^4 - 16}$$

Solution:

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{x^5 - 32}{x^4 - 16} &= \frac{5}{4} (2)^{5-4} \\ &= \frac{5}{4} (2) \\ &= \frac{5}{2}\end{aligned}$$

Therefore:

$$\frac{5}{2}$$

Example 18

Evaluate:

$$\lim_{x \rightarrow 2} \frac{x^6 - 64}{x^3 - 8}$$

Solution:

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{x^6 - 64}{x^3 - 8} &= \frac{6}{3} (2)^{6-3} \\ &= 2(2^3) \\ &= 16\end{aligned}$$

Therefore:

$$16$$

Example 19

Evaluate:

$$\lim_{x \rightarrow -3} \frac{x^5 + 243}{x^4 - 81}$$

Solution:

$$\begin{aligned}\lim_{x \rightarrow -3} \frac{x^5 + 243}{x^4 - 81} &= \lim_{x \rightarrow -3} \frac{x^5 - (-3)^5}{x^4 - (-3)^4} \\ &= \frac{5}{4} (-3)^{5-4} \\ &= \frac{5}{4} (-3) \\ &= -\frac{15}{4}\end{aligned}$$

Therefore:

$$-\frac{15}{4}$$

Example 20

Evaluate:

$$\lim_{x \rightarrow 2} \frac{x^7 - 128}{x - 2}$$

Solution:

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{x^7 - 128}{x - 2} &= \frac{7}{1} (2)^{7-1} \\ &= 7(2^6) \\ &= 448\end{aligned}$$

Therefore:

$$448$$